

Machine Learning

13. Ensemble Learning

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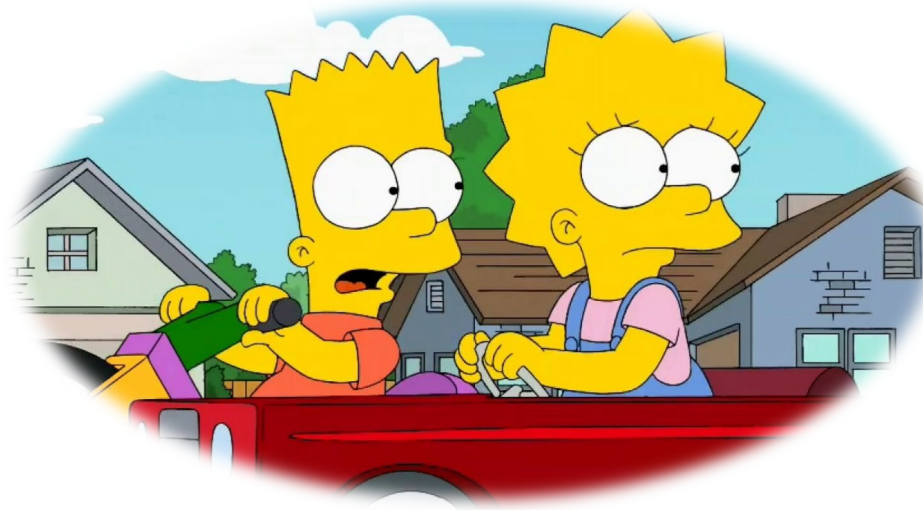
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Today, we will learn...

Ensemble: Can we make multiple models work together?



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- ***Basic Concept of Ensemble Learning***
- Ensemble Methods

How Could We Use Multiple Models at Once?

다른 데이터를 사용한 같은 모델 또는 같은 데이터를 사용한 다른 모델을 사용

- So far, we have considered 'one model for one problem' cases.
- Why not use multiple models to tackle a single problem?
- Can we always expect better predictions by using multiple model?



Practical Application: Netflix Rating Prize

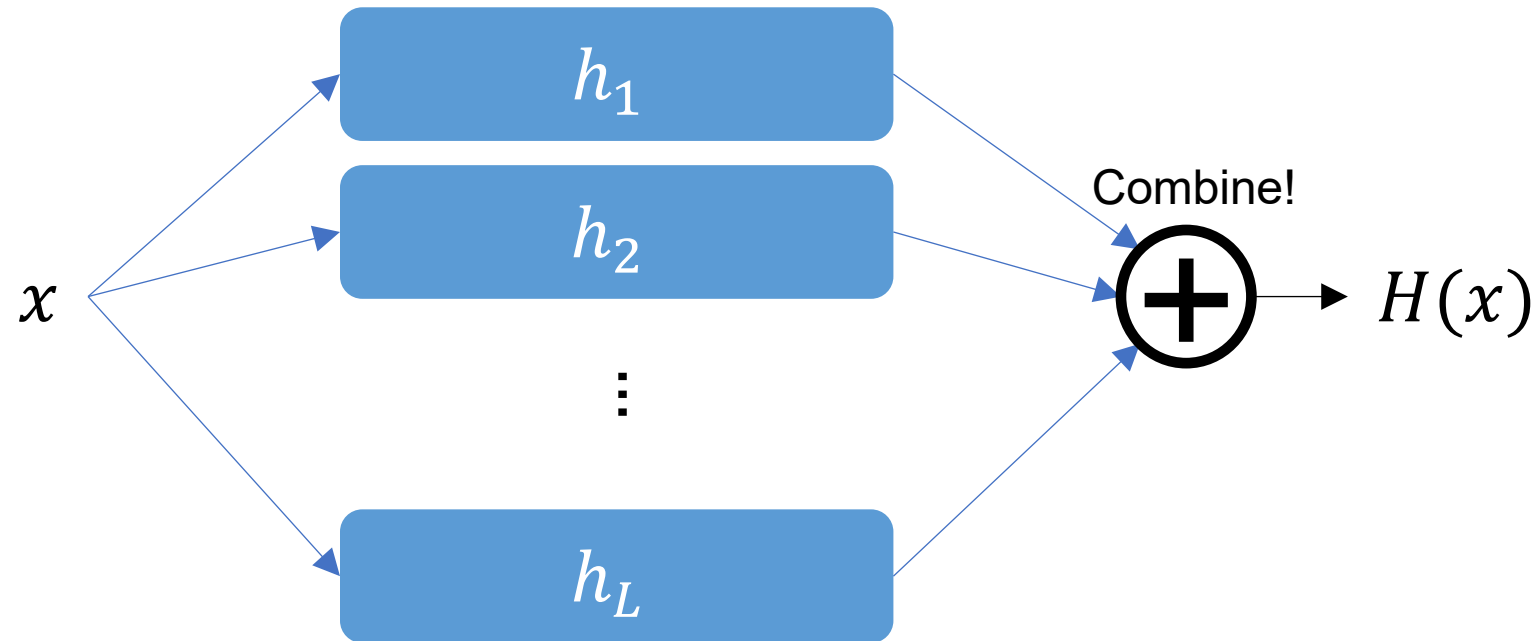
- Let us build up a model that predicts a user's rate on a specific movie.
 - We can think of two factors:
 - The target user's ratings on other movies.
 - Other users' ratings on the target movie.
- } Collaborative filtering



- Netflix Prize: \$1,000,000 to the first team to do 10% better than Netflix system (2007~2009)
- Winner: Bellkor's Pragmatic Chaos: an **ensemble of more than 800 rating systems.**

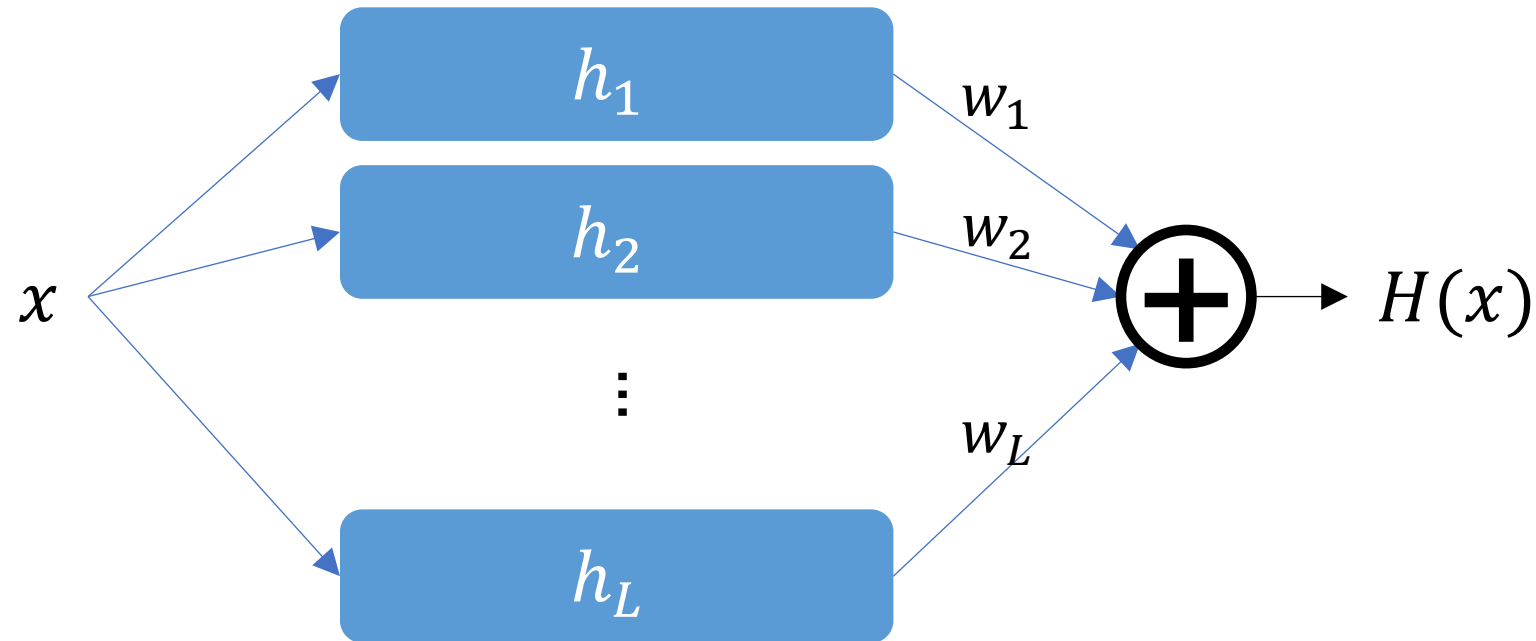
Possible Concepts (1/4)

- Combine classifiers.
- The final hypothesis is just a simple '**vote**' of the members.



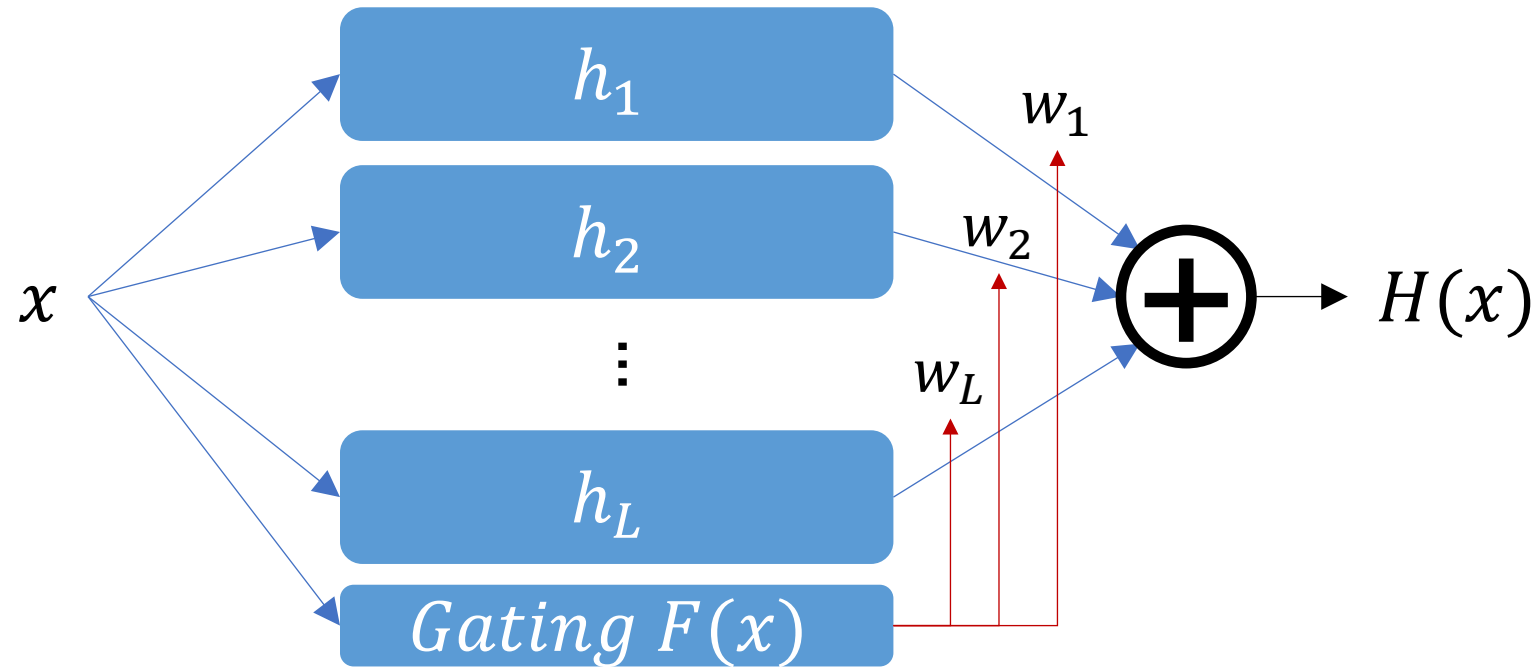
Possible Concepts (2/4)

- The final hypothesis is a ***weighted average*** of multiple classifiers.
- Coefficients can be trained using a validation set.



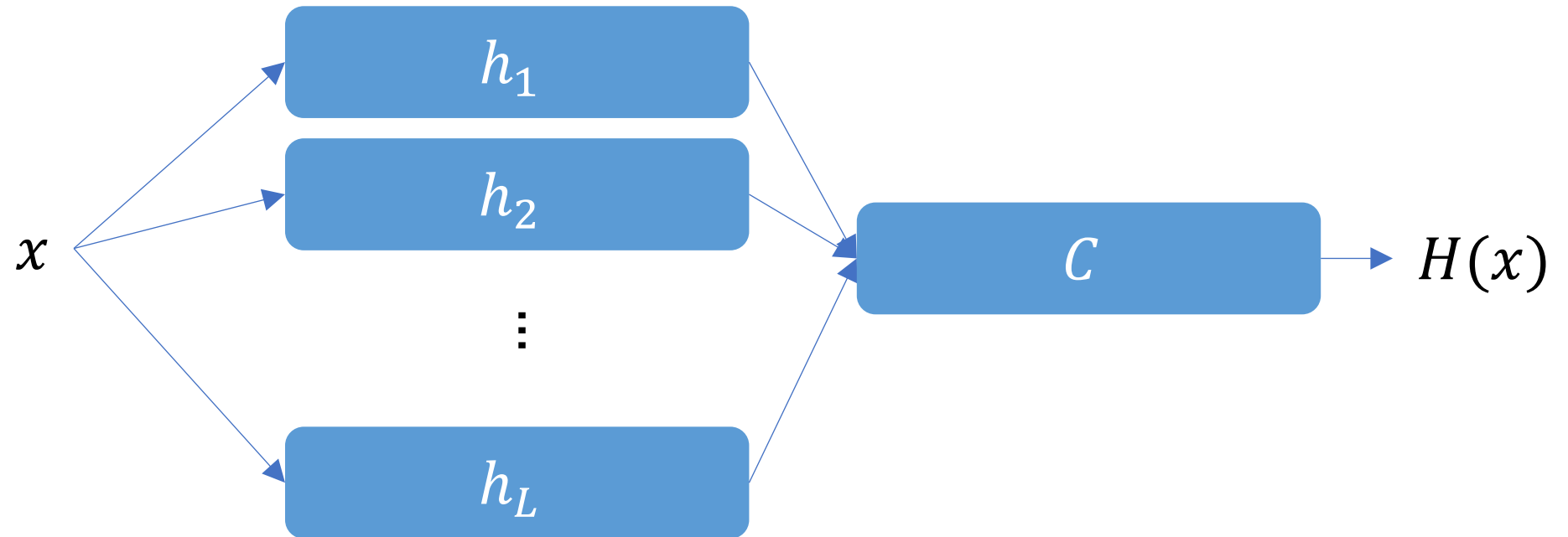
Possible Concepts (3/4)

- **Gating:** An additional function is added between the input and the weights.
- The weights are dependent on not only the validation set but also the input data.



Possible Concepts (4/4)

- **Stacking**: Have an additional classifier that receives the predicted hypothesis from all classifiers as the input.



Diversity Matters!

How to Achieve Diversity

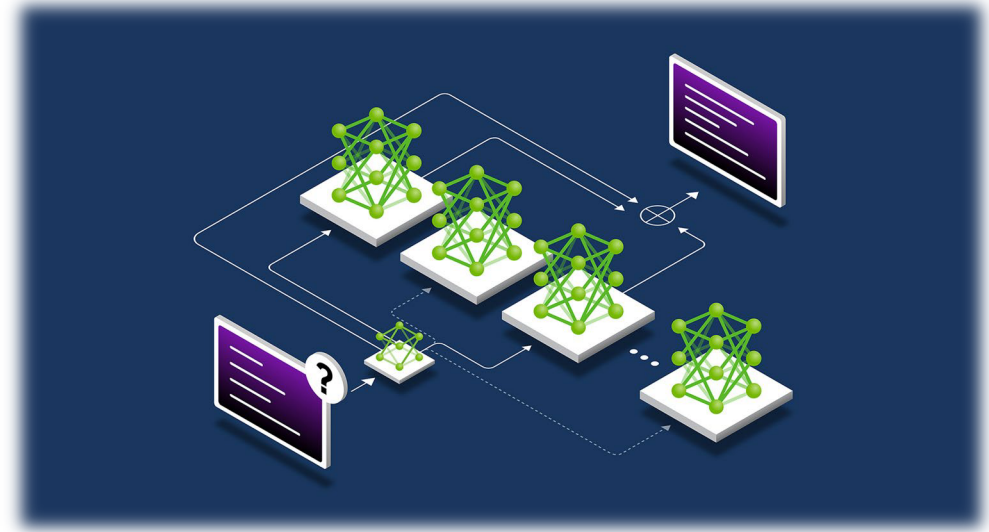


서로 다른 모델간의 오류 패턴의 다양성, 모델간의
오류 패턴이 서로 달라야 앙상블의 의미가 있음

Cause of Mistakes	Diversification Strategy
Pattern was difficult	...
Overfitting	Vary the training datasets
Some features are noisy	Vary the set of input features

Modern Examples of Diversity (1/2)

- Mixture-of-Experts (MoE)
 - Switch Transformer (Google, 2021)
 - GShard (Google, 2020)
 - Mixtral (Mistral AI, 2024)



- As more layers trained on different latent spaces are mixed, the model can better utilize its learning capacity.

여러 전문가 모델 중 입력에 따라 일부 모델만 활성화하여 사용,
하나의 거대한 모델이 모든 문제를 해결하지 않아도 됨

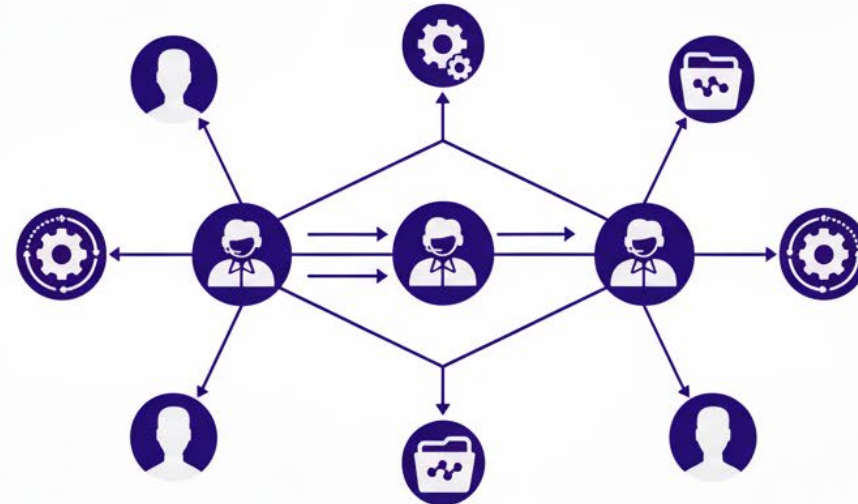
Modern Examples of Diversity (2/2)

- Multi-Agent Systems

Single Agent System

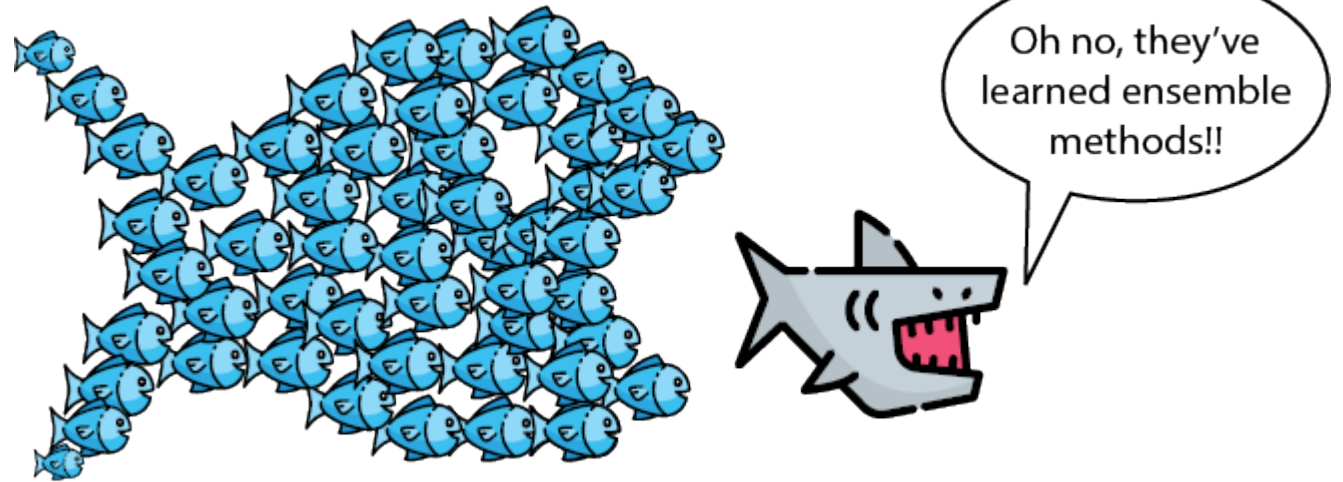


Multi Agent System



Ensemble Methods

- Ensemble is a fundamental machine learning paradigm where multiple models (often called “weak learners”) are trained to solve the same problem and combined to get better results.
- Three representative ways to improve diversity.
 - Bagging
 - Boosting
 - Stacking



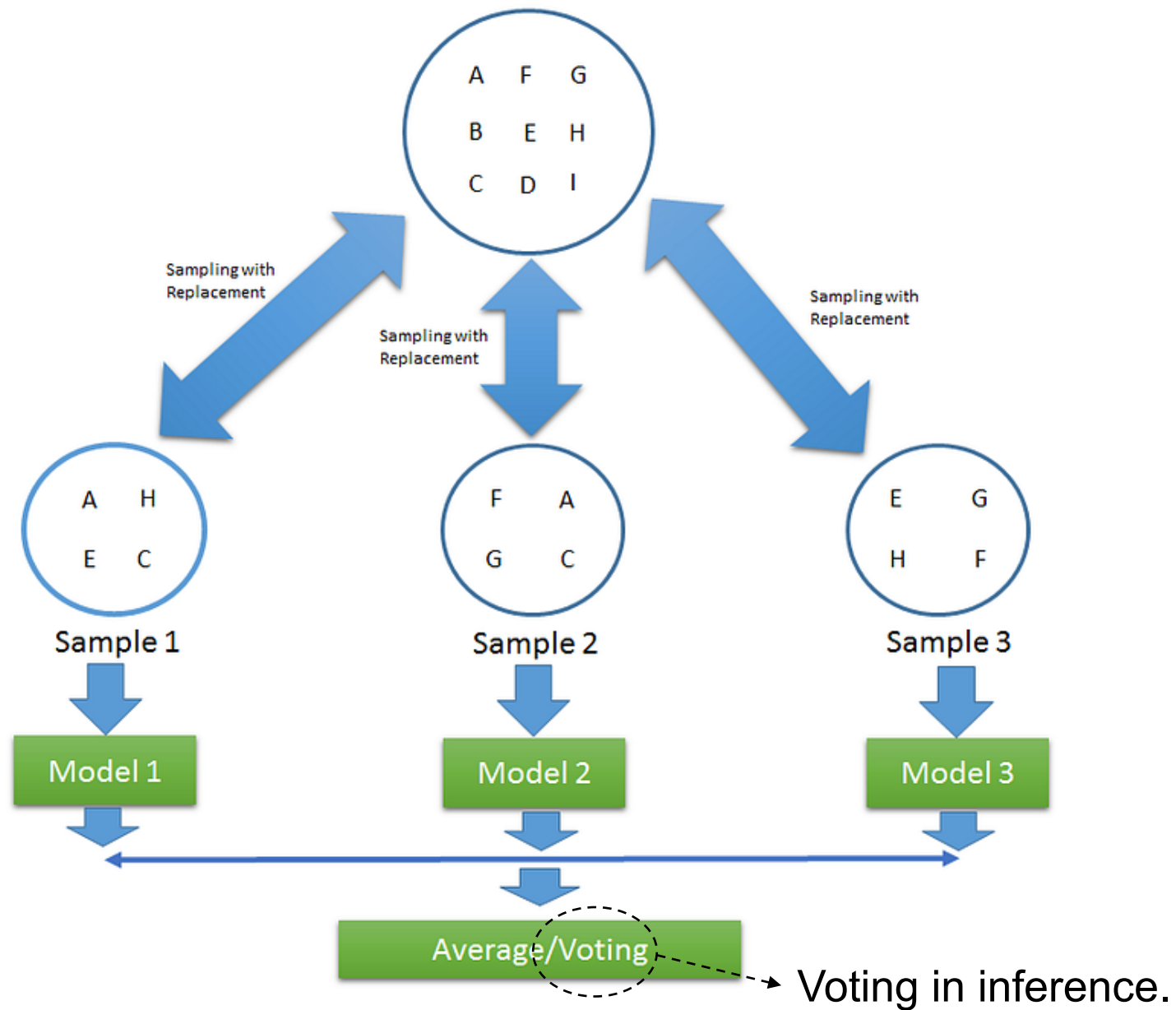
- Basic Concept of Ensemble Learning
- ***Ensemble Methods***
 - ***Bagging***
 - Boosting
 - Stacking

Bagging

데이터를 복원 추출하여 K개의 데이터 군집 생성, 같은 종류의 모델에 학습시켜 각각의 결과를 집계

- Bootstrap Aggregation (Bagging)
 - A technique of sampling different sets of data from a given training dataset with replacement.
 - E.g., randomly sample 70% of the training dataset k times.
 - Once the data is split, ***train k independent models*** on the sub-datasets and ***aggregate*** the result.
- Key Questions
 - How many subsets?
 - How to aggregate the k results.





Benefits of Bagging

- ***Simple implementation***: bagging generally employs the same model for all classifiers.
- ***Better generalization***: Since each weak learner views only a subset of the dataset, it leads to a higher noise scale, and thus results in improving generalization performance.
- ***Parallelization***: All weak learners are basically independent of each other (embarrassingly parallel).



모델의 훈련 성능 자체의 개선은
없으나 일반화 성능은 개선됨

Challenges of Bagging

- ***Loss of Interpretability***: Because multiple weak classifiers are just averaged, it is usually not allowed to figure out the internal mechanism of making better predictions.
- ***Computationally Expensive***: Unless you have large-scale distributed systems, you should sequentially train multiple models!
- ***Less Flexible***: If the data contains a large degree of inherent variance, bagging does not provide much benefits.



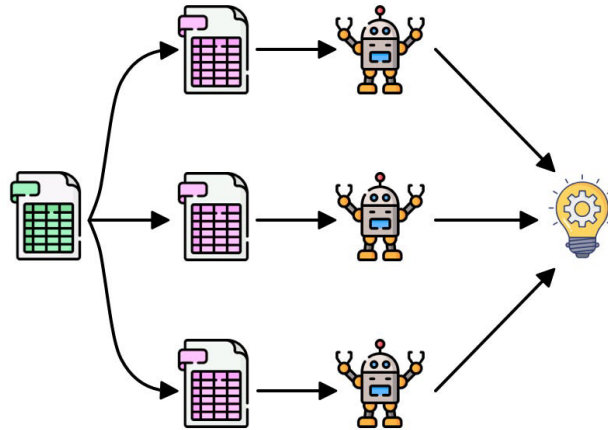
- Basic Concept of Ensemble Learning
- ***Ensemble Methods***
 - Bagging
 - ***Boosting***
 - Stacking

Bagging vs. Boosting

- Instead of multiple independent weak learners, use them sequentially to boost up what they learn!

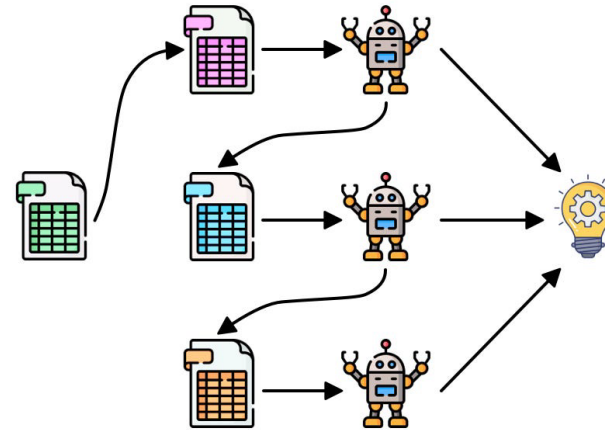
랜덤 데이터셋으로 학습을 시작하되 이전 단계 모델의 오답을 다음 모델이 학습, 이를 제뇌하면 배깅과 동일

Bagging



Parallel

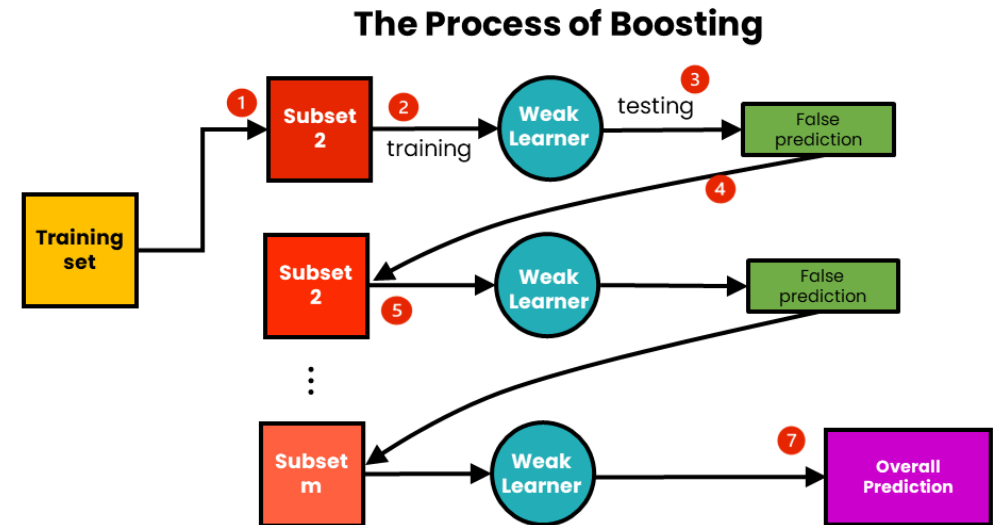
Boosting



Sequential

Boosting

1. We sample k subsets from an initial training dataset.
2. Using the first subset, we train the first weak learner.
3. We test the trained weak learner using the training data. As a result of the testing, **some data points will be incorrectly predicted.**
4. Each **data point with the wrong prediction is sent into the second subset of data**, and this subset is updated.
5. Using this updated subset, we **train and test the second weak learner.**
6. We **continue with the following subset until the total number of subsets is reached.**
7. We now have the total prediction. The overall prediction has already been aggregated at each step, so there is no need to calculate it.



Benefits of Boosting

- **Easy Implementation:** Just a cascaded training of multiple models.
바닐라 모델에 비해서는 과적합 성능이 낮지만 배깅과 비교해서는 과적합 가능성 발생
- **Less Overfitting:** Due to the sequential connections, the model complexity is increased as having more weak learners.
- **Computational Efficiency:** Since each weak learner focuses on what the previous learner failed to learn, the model complexity can be relatively low → cheaper training cost.



Challenges of Boosting

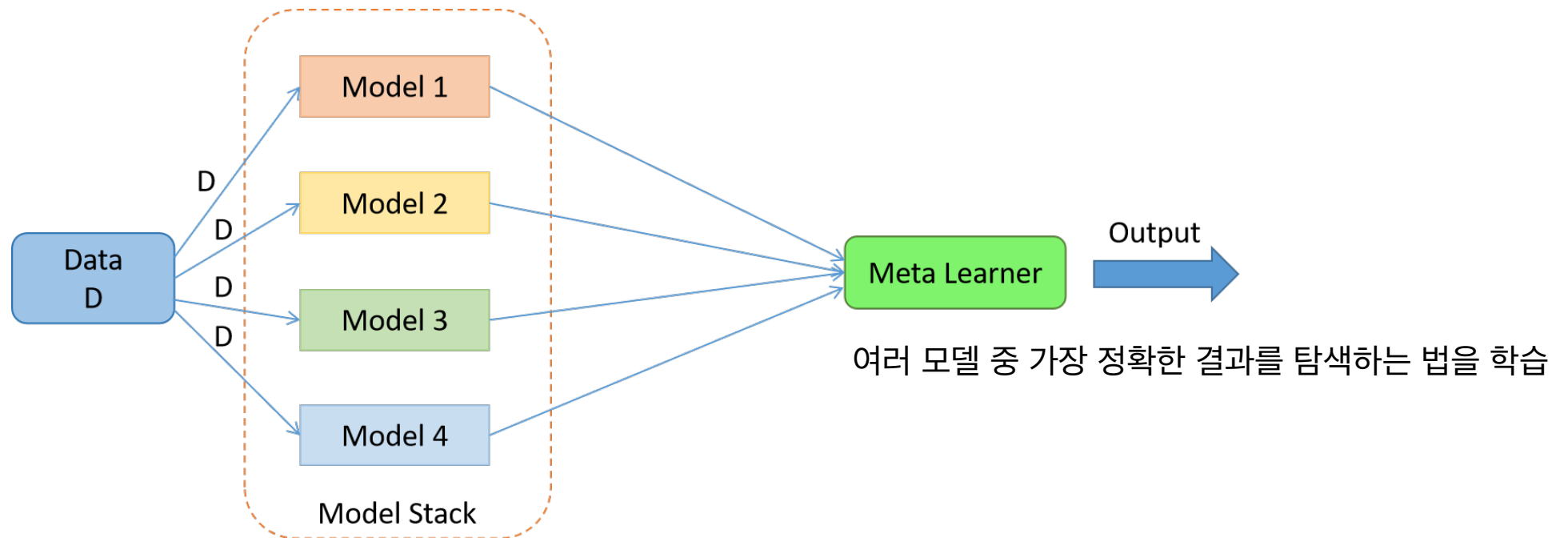
- **Overfitting:** There are disputes in research around whether boosting alleviates or exacerbates the overfitting. Nevertheless, in some cases, it can cause overfitting.
- **Scalability:** Sequential connections introduce the data dependency, and it does not allow to parallelize the training of k weak learners.



- Basic Concept of Ensemble Learning
- ***Ensemble Methods***
 - Bagging
 - Boosting
 - ***Stacking***

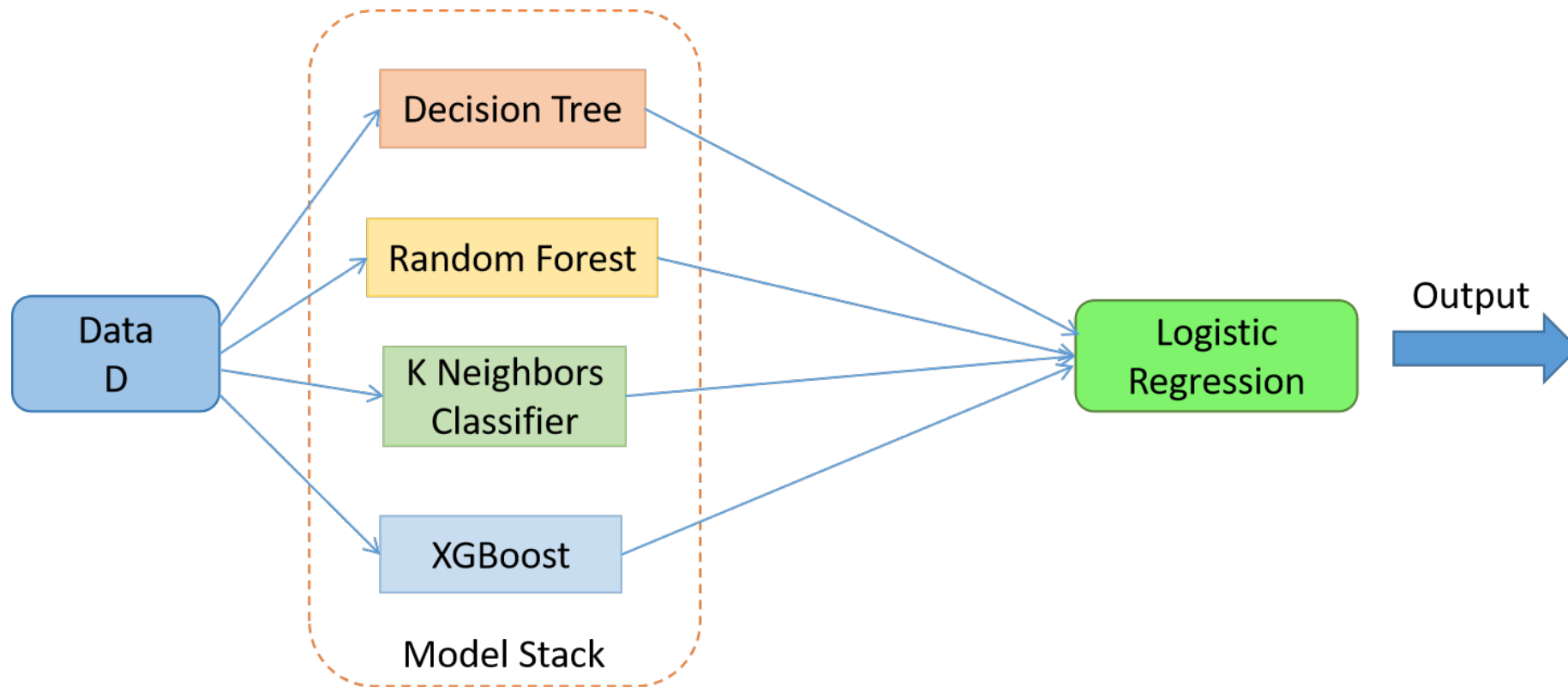
Can We A Bit More Generalize Bagging?

- Stacking is a generalized version of bagging.
- Instead of just simply aggregating the weak learners' predictions, train an additional model!



Practical Stacking

Different types of models can be stacked!



Okay Then, What Should Be the Meta Model?

- In many existing research works, the meta-model tends to be a simple ***linear regression model for regression tasks*** and a ***logistic regression model for classification tasks***.
- Why? Because the meta model only views the output predictions rather than the comprehensive input information, it is easy to overfit to the given input predictions.

모델의 출력이 메타러너의 입력이 되므로, 메타러너의 입력의 공간 크기는 매우 작음
이진 분류 모델의 출력을 메타러너의 입력으로 사용하면 2차원 데이터만 입력됨

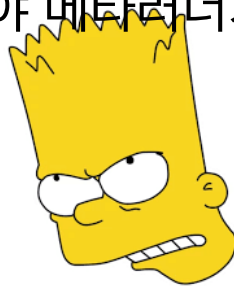
Benefits of Stacking

- ***Robust Predictions***: The meta model likely makes more robust predictions than any individual models because it views many possible false predictions. 여러 개의 서로 다른 모델의 예측을 조합, 개별 모델 하나에 의존할 때보다 더 안정적이고 일반화 성능이 개선됨, 분산/편향/과적합이 감소함
- ***Ease of Implementation***: Just a concatenation of a set of weak learners and a single meta model.



Challenges in Stacking

- **Expensive Computational Cost:** As compared to bagging, it requires an extra model (meta learner) to be trained.
- **Sensitive to Diversity:** If there is any strong correlation across weak learners, the meta model just learns simple combinations of the similar predictions. 다양성이 낮으면 사실상 보팅 기법과 차이가 없어짐
- **Scalability:** Due to the data dependency between weak learners and the meta learner, their training should be sequential. 각 모델 학습이 진행되어야 메타러너가 동작하므로 병렬화 어려움



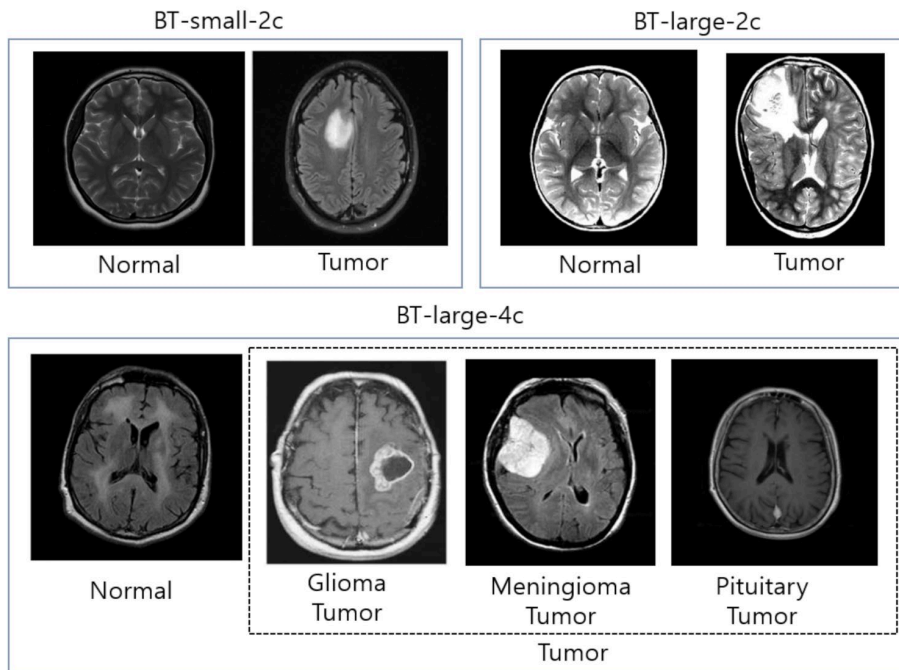
Ensemble Learning Summary

- In general, ensemble methods are expected to yield better predictive performance than a single model.
- However, you should train multiple models, having more expensive computational cost.
- Nevertheless, ensemble is an effective way of making synergetic performance with multiple learners!

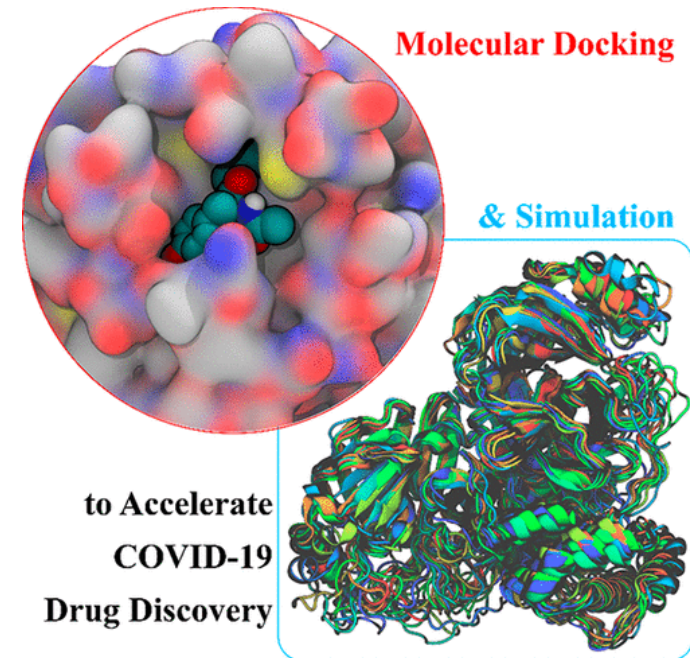
모델 성능을 획기적으로 개선시키지는 못 하나, 일반화 성능을 간단한 구현과 비용으로 개선시킬 수 있는 트릭의 역할

Real-World Use-Cases

MRI-Based Brain Tumor
Classification Using Ensemble of
Deep Features and ML Classifiers.

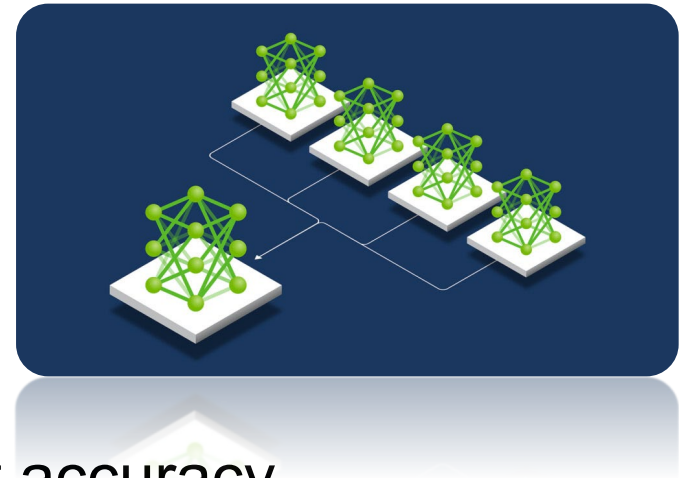


Ensemble Docking Drug Discovery
Pipeline with Application to Covid-19

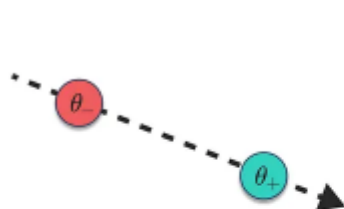


Over the Simple Ensemble...

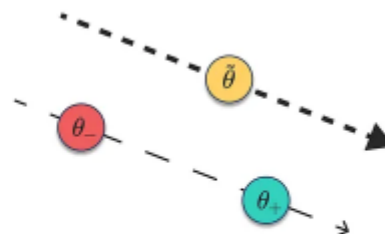
- Model Merging
 - A popular research topic as of late, but the history goes all the way back to the mid 1990s.
- Mode Connectivity
 - Some researchers found out that the weights of neural networks independently trained can be connected together in the loss landscape via constant training and test accuracy.



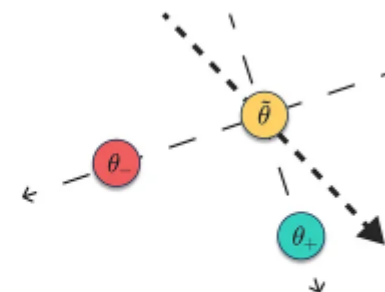
Model Merging



Interpolation between two fine-tuned parameters

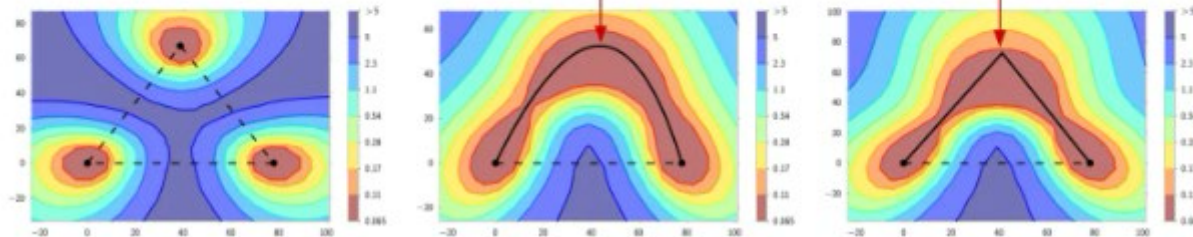


Moving from a pre-trained state in the direction obtained by two fine-tuned parameters



Moving from a pre-trained state with basis formed by fine-tuned parameters

Two modes are connected via a simple path / curve



Model Souping

- Google Brain[1] released a simple but effective ML methodology, model soup.
- When fine-tuning pre-trained models, just simple averaging of multiple independent models yields a superior model!

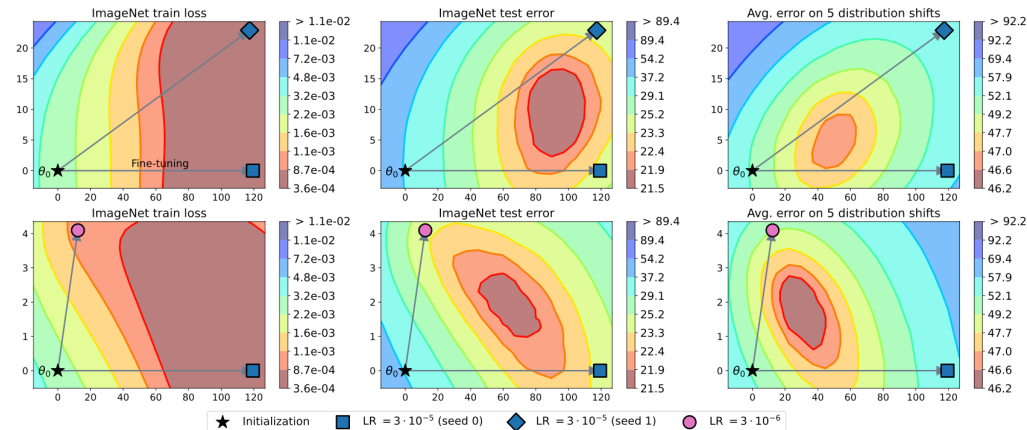


Figure 2: The solution with the highest accuracy is often not a fine-tuned model but rather lies between fine-tuned models. This figure shows loss and error on a two dimensional slice of the loss and error landscapes. We use the zero-shot initialization θ_0 and fine-tune twice (illustrated by the gray arrows), independently, to obtain solutions θ_1 and θ_2 . As in Garipov et al. (2018), we obtain an orthonormal basis u_1, u_2 for the plane spanned by these models, and the x and y -axis show movement in parameter space in these directions, respectively.

Any Questions?

Good work. We have finished all the topics for this semester.
Hope you all enjoyed what have been covered so far😊

